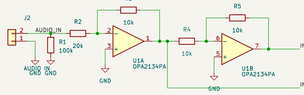
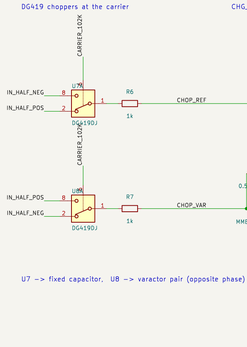
1 INPUT

phase splitter: IN_HALF_NEG = −x/2, IN_HALF_POS = +x/2



3 MODULATOR

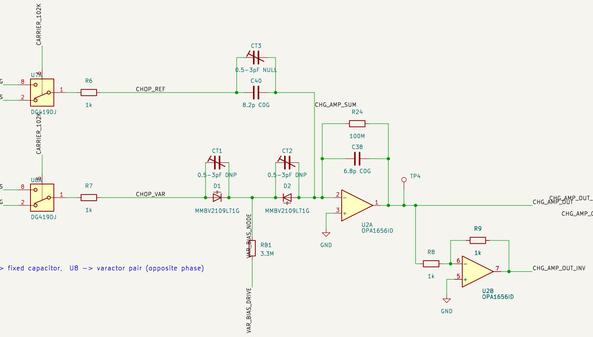
DG419 choppers at the carrier



U7 −> fixed capacitor, U8 −> varactor pair (opposite phase)

4 HALF-BRIDGE + CHARGE AMP

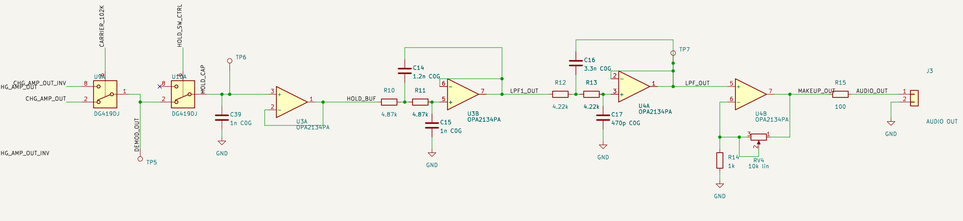
CHG_AMP_SUM and VAR_BIAS_NODE are high-impedance: PTFE standoffs, no breadboard



Gain = (C_pair − C_fixed) / (2 Cf). CT3 sets the null near VB = 12 V. CT1/CT2 optional pair-match trim.

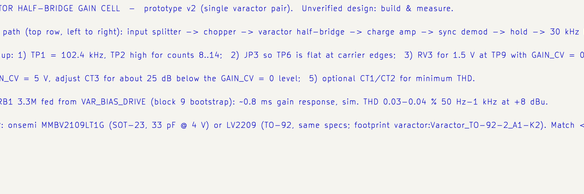
5 DEMOD, HOLD, FILTER, OUTPUT

sync demod + gated hold + 30 kHz 4th-order Butterworth + makeup gain



2 TIMING

3.2768 MHz / 32 = 102.4 kHz carrier; HOLD_GATE high for counts 8..14 of each half-period



JP3: choose the position where TP6 (HOLD_CAP) is a clean staircase with no spikes at carrier edges

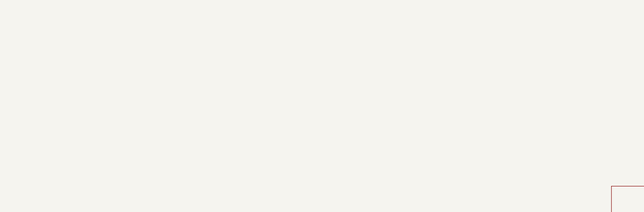
6 CONTROL

VAR_BIAS_CMD = 2 x GAIN_CV + 2 x VB_OFFSET (VB_OFFSET −0.75 V: 1.5..11.5 V for GAIN_CV 0..5 V)

RV3: set VAR_BIAS_CMD = 1.5 V with GAIN_CV = 0. JP1 selects external CV (J4) or the manual pot RV2.

9 BIAS SERVO

bootstrapped feed: VAR_BIAS_DRIVE = VAR_BIAS_CMD + 0.99 x HP1Hz(BIAS_SENSE − REP_SENSE)



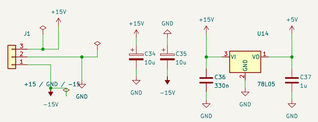
Guard ring around VAR_BIAS_NODE (D1/D2 cathodes, RB1, U16A pin 3) driven by BIAS_SENSE. Replica D3/D4: matched to D1/D2 within 0.2 %, same thermal block.

Trim RV5: no audio, CV = 0.5 Hz square (small steps). Adjust RV5 until TP11 (BIAS_ERR) shows no step-shaped transients.

U17/U18: INA128 at G = 1 (RG open). U18 REF carries VAR_BIAS_CMD with unity gain to the drive − this exactness is what keeps gain moves clean.

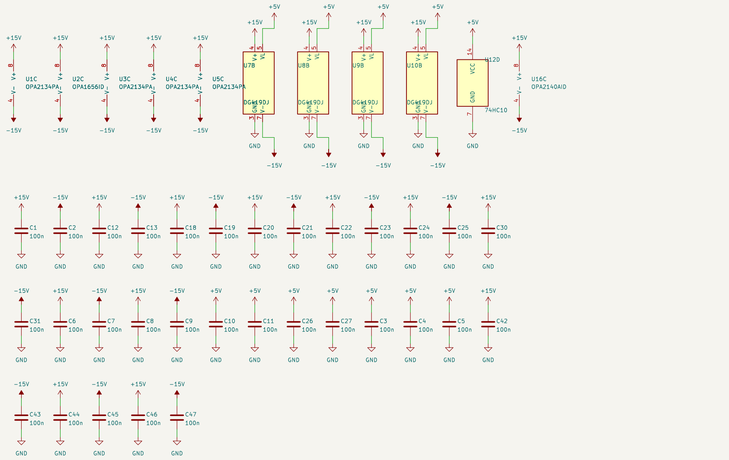
7 POWER

+/−15 V in, 78L05 for logic and DG419 VL



8 IC SUPPLY PINS + DECOUPLING

one 100n per supply pin, placed right at the IC



C1,C2: U1 C12,C13: U2 C18–C21: U9,U10 C22–C25: U3,U4 C30,C31: U5 C6–C9: U7,U8 C10,C11,C26,C27: DG419 VL C3–C5: X1, U11, U12 C42,C43: U16 C44,C45: U17 C46,C47: U18

VARACTOR HALF-BRIDGE GAIN CELL — prototype v2 (single varactor pair). Unverified design: build & measure.

Signal path (top row, left to right): input splitter −> chopper −> varactor half-bridge −> charge amp −> sync demod −> hold −> 30 kHz LPF −> makeup gain.

Bring-up: 1) TP1 = 102.4 kHz, TP2 high for counts 8..14; 2) JP3 so TP6 is flat at carrier edges; 3) RV3 for 1.5 V at TP9 with GAIN_CV = 0;

4) GAIN_CV = 5 V, adjust CT3 for about 25 dB below the GAIN_CV = 0 level; 5) optional CT1/CT2 for minimum THD.

Bias: RB1 3.3M fed from VAR_BIAS_DRIVE (block 9 bootstrap): −0.8 ms gain response, sim. THD 0.03–0.04 % 50 Hz–1 kHz at +8 dBu.

D1, D2: onsemi MMBV2109LT1G (SOT-23, 33 pF @ 4 V) or LV2209 (TO-92, same specs; footprint varactor:Varactor_TO-92-2_A1-K2). Match <1 %, thermally coupled.

One anti-series varactor pair vs fixed C0G; carrier-fed; charge amp + sync demod. 0 to −25 dB pe

Sheet: /
File: varactor_halfbridge.kicad_sch

Title: Varactor half-bridge gain cell — v3 (bootstrapped bias servo)

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